

RSM338 Lab Report #2

February 7, 2026

#Problem 1 This section conducts a preliminary analysis to understand four objective risk indicators' distributions and interrelationship across 122 countries. The goal is to recognise strong correlations between risk features and potential outliers with the justification of why standardization is needed before applying unsupervised clustering algorithms.

#Task(a) Exploratory Data Analysis In this section, I create a scatter plot matrix that shows the relationships between all pairs of features, including Corruption, Peace, Legal, and GDP Growth. This helps to identify correlations, outliers, and potential groupings within the country risk data.

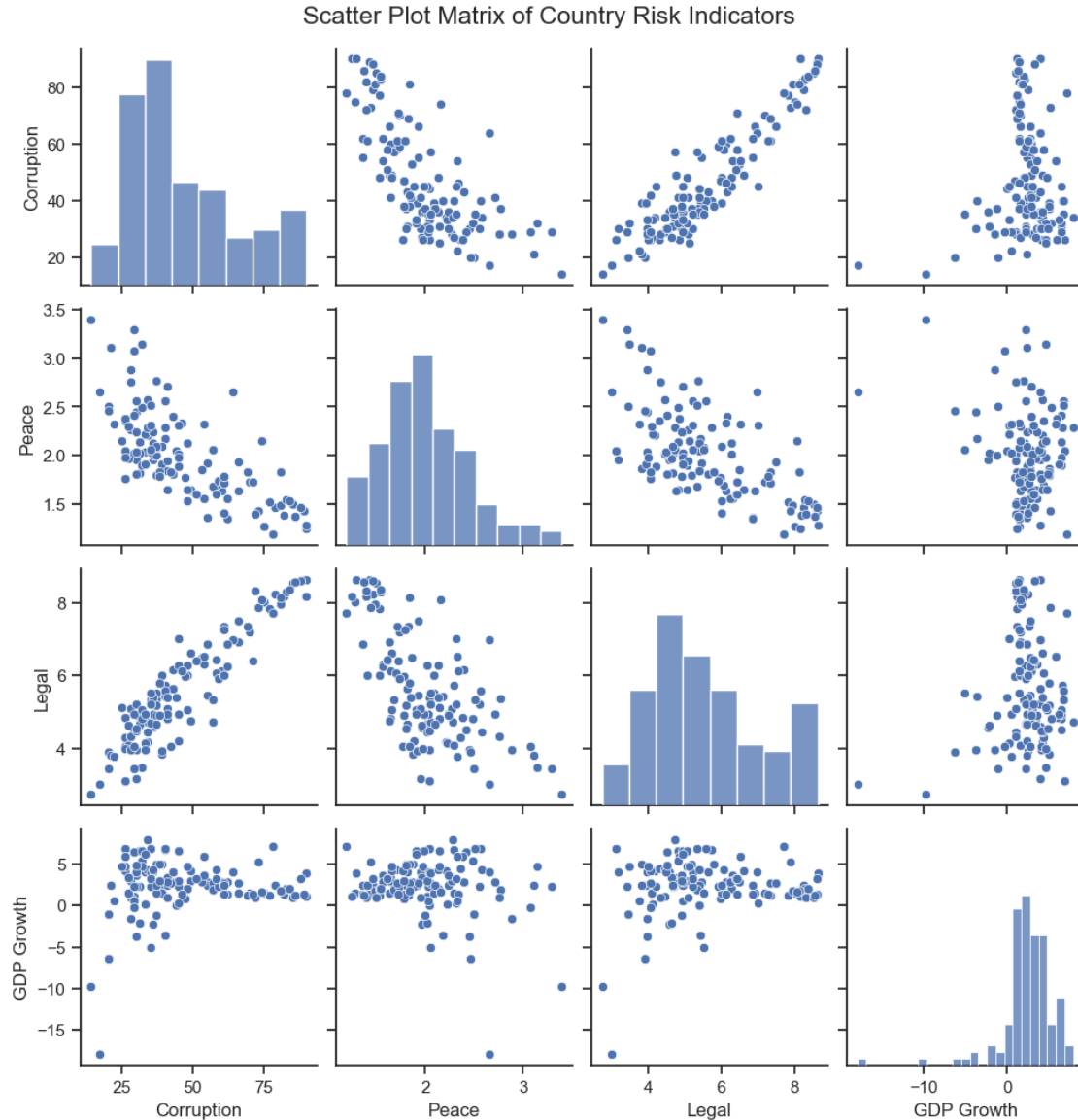
```
[3]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# 1. Load data
df = pd.read_csv('countryriskdata.csv')

# 2. Extract only the numeric features for the pairplot
features = ['Corruption', 'Peace', 'Legal', 'GDP Growth']

# 3. Create the scatter plot matrix
sns.set_theme(style="ticks")
pair_plot = sns.pairplot(df[features], diag_kind='hist')

pair_plot.fig.suptitle("Scatter Plot Matrix of Country Risk Indicators", y=1.02, fontsize=16)
plt.show()
```



There is a strong positive linear correlation between Corruption and Legal. This means that countries with higher transparency show higher property rights protection. In addition, Peace with both Corruption and Legal also show notable negative correlations, indicating that more peaceful countries would have better legal and transparency ratings. There are two main groupings that are visible in the Corruption and Legal plot. One distinct cluster of countries can be observed in the top-right corner of the Corruption vs Legal plot, demonstrating highly stable and low risk nations. The remaining countries are more spread out in the lower-left and middle sections of the plots, having more varied risk profiles. Those scatter plots which include GDP Growth have several outliers as there are 1 or 2 countries that sit very far away from the main group. Most countries are grouped between 0% and 5% growth, however, there are a few countries which are grouped below -10% or above 5% that do not seem strongly correlated.

#Task(b) Correlation Maxtrix of country risk indicators

```
[5]: # 1. Compute the correlation matrix
correlation_matrix = df[features].corr()

# 2. Display the correlation matrix as a table
print("--- Correlation Matrix: Country Risk Features ---")
display(correlation_matrix.round(4))
```

```
--- Correlation Matrix: Country Risk Features ---
```

	Corruption	Peace	Legal	GDP Growth
Corruption	1.0000	-0.7005	0.9236	0.1007
Peace	-0.7005	1.0000	-0.6520	-0.2006
Legal	0.9236	-0.6520	1.0000	0.1183
GDP Growth	0.1007	-0.2006	0.1183	1.0000

According to the correlation matrix, Corruption and Legal pair have the strongest correlation. This makes intuitive sense from a country risk perspective because the Corruption index measures transparency, while Legal risk index measures property rights protection. If the government is “clean” and transparent, strong legal institutions and the protection of property rights can be maintained. However, a lack of protection can create an environment where corruption may grow. For an investor, this suggests that legal stability and government transparency are not independent risks but are part of one single “Institutional Strength” dimension.

#Task(c) The necessity of standardization Standardization is required before clustering because all four features have different units. For example, Corruption is measured ranges from 0 to 100, while Peace has ranges from 1 to 5. Hence, standardization is used for transforming these into the same scale so they all have a mean of 0 and a standard deviation of 1. If we ran K-Means on the unstandardized data, the algorithm would be misled by the size of the numbers rather than the actual patterns in the data. In this given dataset, Corruption would dominate the distance calculations because it has the largest numerical range. Corruption’s values vary up to 100, while features such as Legal only increase up to 10. Algorithms may recognize small changes in corruption as more important than large changes in other features. Hence, this can cause the results of ignoring other three features and are only grouped by Corruption.

RSM338 Lab Report #2 Problem 2

February 7, 2026

#Problem 2 In this section, K-Means clustering algorithm is applied to categorize 122 countries into distinct risk profiles based on four features. Elbow method and Silhouette analysis are utilized to determine the optimal number of clusters, ensuring a balance between mathematical precision and practical business utility.

Data preparation In this section, I extract the four numeric features and apply standardization. This ensures that each indicator which is Corruption, Peace, Legal, and GDP Growth contributes equally to the distance used by the K-Means algorithm.

```
[1]: import pandas as pd
from sklearn.preprocessing import StandardScaler

# Load the data
df = pd.read_csv('countryriskdata.csv')

# Extract the four numeric features into a feature matrix X
features = ['Corruption', 'Peace', 'Legal', 'GDP Growth']
X = df[features]

# Standardize the features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Create a DataFrame to view the standardized data
X_scaled_df = pd.DataFrame(X_scaled, columns=features)

print("--- Data Preparation Complete ---")
display(X_scaled_df.head())
```

--- Data Preparation Complete ---

	Corruption	Peace	Legal	GDP Growth
0	-0.379975	-0.306578	-1.199593	0.808020
1	-0.642472	0.469231	-0.971402	0.561381
2	-0.537473	-0.104778	-0.695954	-1.447586
3	-0.694971	0.480442	-0.994356	-0.673052
4	1.720000	-1.207951	1.785784	0.026378

#Task(a) The Elbow Method for optimal K In this section, the Elbow method is applied to find the point where adding more clusters no longer significantly improves our data.

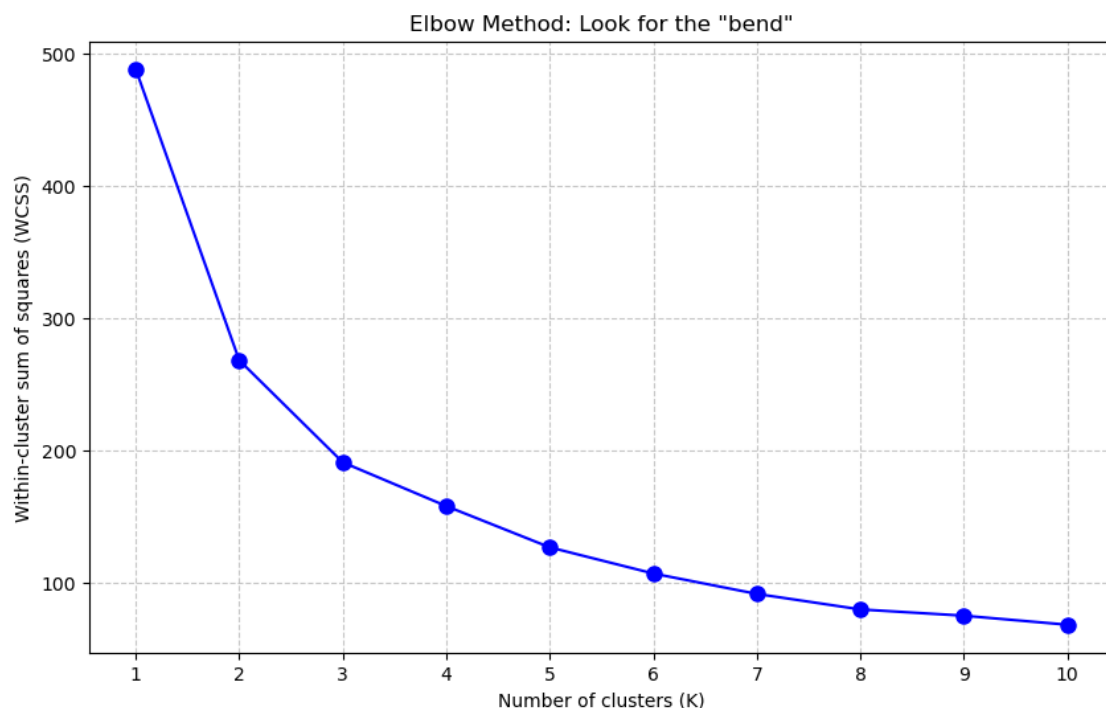
```
[3]: import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

# Compute WCSS for K = 1 through 10
wcss = []
K_range = range(1, 11)

for k in K_range:
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(X_scaled) # Use your standardized data here
    wcss.append(kmeans.inertia_)

# Plotting the results
plt.figure(figsize=(10, 6))
plt.plot(K_range, wcss, 'bo-', markersize=8)
plt.xlabel('Number of clusters (K)')
plt.ylabel('Within-cluster sum of squares (WCSS)')
plt.title('Elbow Method: Look for the "bend"')
plt.xticks(K_range)
plt.grid(True, linestyle='--', alpha=0.7)

plt.show()
```



Based on the “elbow” in this plot, I would recommend 3 clusters ($K=3$). This is because before

the elbow, there is a very steep drop in WCSS. After adding the second and third clusters, the within-cluster sum of squares reduced substantially. This indicates that moving from 1 to 3 clusters captures a massive amount of information about the differences between countries. At K=3, the bend is most prominent and the slope changes significantly. Adding a 4th or 5th cluster provides diminishing returns and the line becomes much flatter. This shows that the improvement in cluster tightness is limited compared to the added complexity of the model.

#Task(b) Running the K-Means Model

```
[5]: from sklearn.cluster import KMeans
import pandas as pd

# Fit the K-Means model with K=3
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
kmeans.fit(X_scaled)

# Extract the cluster centers
centroids = kmeans.cluster_centers_

# Create the table with one row per cluster and one column per feature
features = ['Corruption', 'Peace', 'Legal', 'GDP Growth']
centroids_df = pd.DataFrame(centroids, columns=features)
centroids_df.index.name = 'Cluster'

print("--- Cluster Centers (Standardized Units) ---")
display(centroids_df.round(4))
```

```
--- Cluster Centers (Standardized Units) ---

      Corruption   Peace   Legal  GDP Growth
Cluster
0          -0.4796  0.2633 -0.4351      0.3759
1           1.2314 -0.9792  1.1660      0.0062
2          -1.0275  1.3522 -1.0590     -1.7201
```

Since the data is standardized, a positive value means the cluster is above average for that feature, and a negative value means it is below average. Cluster 0 can be explained as the group with moderate risk and emerging growth. It has both negative standardized scores of Corruption and Legal, while has the highest scores of GDP Growth. This suggests that developing countries can be categorized under this cluster as they are growing faster than average despite having some institutional weaknesses or moderate corruption. Cluster 1 has the highest corruption and legal scores. It also has a negative Peace score which indicates countries are more peaceful than average. Overall, these countries represent the safest investment environment because they are stable, democratic, and highly transparent. Cluster 2 has the worst scores across different features as it reflects the lowest transparency, lowest legal protection and lowest GDP Growth. These countries are facing economic contraction alongside systemic instability. They are highly volatile and have high risk environments which hinder economic growth.

#Task(c) Mapping Clusters to Countries

```
[7]: # Fit the model
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
kmeans.fit(X_scaled)

# Assign the cluster labels to a new column in your dataframe
df['Cluster'] = kmeans.labels_

for cluster_id in sorted(df['Cluster'].unique()):
    countries = df[df['Cluster'] == cluster_id]['Country'].values
    print(f"Cluster {cluster_id} Countries ({len(countries)} total):")
    print(f"{'', ' '.join(countries)}\n")
```

Cluster 0 Countries (68 total):

Albania, Algeria, Armenia, Bahrain, Bangladesh, Benin, Bolivia, Bosnia and Herzegovina, Bulgaria, Cameroon, China, Colombia, Croatia, Cyprus, Dominican Republic, Egypt, El Salvador, Ethiopia, Gabon, Georgia, Ghana, Greece, Guatemala, Honduras, India, Indonesia, Iran, Israel, Italy, Jamaica, Jordan, Kazakhstan, Kenya, Kuwait, Liberia, Madagascar, Malawi, Mali, Mauritania, Mexico, Moldova, Montenegro, Morocco, Mozambique, Nepal, Nicaragua, Oman, Pakistan, Panama, Paraguay, Peru, Philippines, Romania, Rwanda, Saudi Arabia, Senegal, Serbia, Sierra Leone, South Africa, Sri Lanka, Tanzania, Thailand, The FYR of Macedonia, Tunisia, Turkey, Uganda, Vietnam, Zambia

Cluster 1 Countries (39 total):

Australia, Austria, Belgium, Botswana, Canada, Chile, Costa Rica, Czech Republic, Denmark, Estonia, Finland, France, Germany, Hungary, Iceland, Ireland, Japan, Korea (South), Latvia, Lithuania, Malaysia, Mauritius, Netherlands, New Zealand, Norway, Poland, Portugal, Qatar, Singapore, Slovakia, Slovenia, Spain, Sweden, Switzerland, Taiwan, United Arab Emirates, United Kingdom, United States, Uruguay

Cluster 2 Countries (15 total):

Argentina, Azerbaijan, Brazil, Burundi, Chad, Democratic Republic of Congo, Ecuador, Lebanon, Nigeria, Russia, Trinidad and Tobago, Ukraine, Venezuela, Yemen, Zimbabwe

The groupings make strong intuitive sense as they categorize countries based on “Risk Profiles”. Cluster 0 includes typical “emerging markets” such as India and Vietnam. This is reasonable because these nations show a healthy GDP Growth, while the institutional score is much lower than cluster 1. These countries are often in transition. Although they are not institutionalized compared to cluster 1, they often see higher GDP Growth as they develop infrastructure and industries. As an investor, seeing India and Vietnam is reasonable as they represent “high risk, high reward”, while needing to embrace some institutional uncertainty.

Cluster 1 includes countries like Australia, Norway, and Singapore. These placements are well justified because they are known for high transparency and strong property rights. Their GDP Growth rates are moderate, but their strong institutional foundations position them as low risk investment environments. Singapore is also known as a country that has a strong rule of law,

leading to build safety and legal certainty over high-risk growth.

Cluster 2 includes countries such as Mali and Venezuela. This is a reasonable placement because these nations have a high risk profile across both political and economic dimensions. A high Peace score and weak institutional quality indicated by low Legal and Corruption support that Mali faces severe internal instability. Venezuela represents economic distress with poor transparency and negative GDP growth. Overall, this cluster has the highest investment risk as it faces political fragility and economic contraction.

#Task(d) Silhouette Analysis

```
[9]: from sklearn.metrics import silhouette_score
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

# 1. Compute silhouette score for K = 2 through 10
silhouette_scores = []
K_range = range(2, 11) # K=1 is not possible for Silhouette

for k in K_range:
    # We keep random_state=42 to match our Elbow Method exactly
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    labels = kmeans.fit_predict(X_scaled)

    score = silhouette_score(X_scaled, labels)
    silhouette_scores.append(score)

# 2. Plotting the results
plt.figure(figsize=(8, 5))
plt.plot(K_range, silhouette_scores, 'go-', markersize=8)
plt.xlabel('Number of clusters (K)')
plt.ylabel('Silhouette Score')
plt.title('Silhouette Analysis')
plt.xticks(K_range)
plt.grid(True, linestyle='--', alpha=0.6)

plt.show()
```




According to Silhouette Analysis, the highest point is at $K=2$ which disagrees with the Elbow method at $K=3$. Silhouette analysis indicates that while binary segmentation is the most mathematically distinct, the elbow method supports that a third cluster still provides important information before reaching diminishing returns. In this context, I would trust the Elbow method. This is because $K=2$ merely categorises the countries into “safe” and “unsafe” which is too simplistic for investment purposes. If we take $K=3$, it provides much higher practical business utility as it identifies a middle segment. There are investors who consider this third group with average institutional stability but high GDP Growth as valuable opportunities. Hence, using $K=3$ from the elbow method is a more meaningful nuance that justifies the slight reduction in clustering accuracy.

RSM338 Lab Report #2 Problems 3 & 4

May 10, 2026

0.1 Problem 3

This problem applies hierarchical clustering to the standardized data, displaying a dendrogram to suggest a natural number of clusters, which we can compare to the results obtained in the K-Means solution.

Data Preparation: In this section, we once again standardized the data to ensure that differences in the scale of measurement of each factor do not impact the clustering decision.

```
[1]: import pandas as pd
      from sklearn.preprocessing import StandardScaler

      # Load data
      df = pd.read_csv("countryriskdata.csv")

      # Feature matrix X (four numeric features)
      features = ["Corruption", "Peace", "Legal", "GDP Growth"]
      X = df[features].values

      # Standardize
      scaler = StandardScaler()
      X_scaled = scaler.fit_transform(X)
      X_scaled = pd.DataFrame(
          scaler.fit_transform(df[features]),
          columns=features
      )
      X_scaled.head()
```

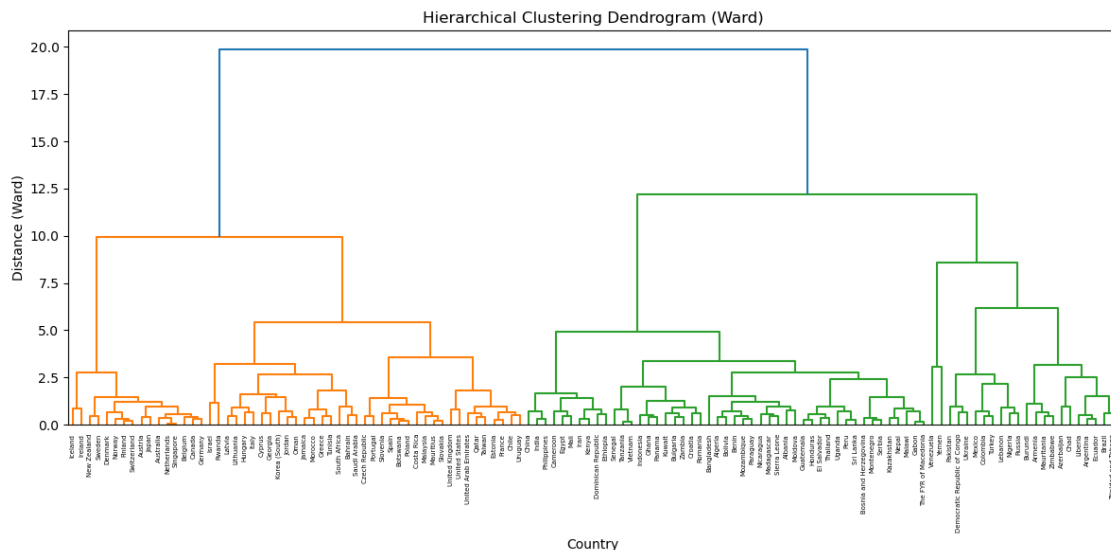
```
[1]:      Corruption      Peace      Legal  GDP Growth
0    -0.379975 -0.306578 -1.199593    0.808020
1    -0.642472  0.469231 -0.971402    0.561381
2    -0.537473 -0.104778 -0.695954   -1.447586
3    -0.694971  0.480442 -0.994356   -0.673052
4     1.720000 -1.207951  1.785784    0.026378
```

Task #3a Agglomerative Hierarchical Clustering using Ward's Linkage: In this section, Ward's linkage is used to perform agglomerative hierarchical clustering on the standardized data and the related dendrogram is displayed to first make a comparison to the K-Means solution and then to discover the natural number of clusters the dendrogram suggests.

```
[2]: from scipy.cluster.hierarchy import dendrogram, linkage
import matplotlib.pyplot as plt

# Ward linkage on standardized data
Z = linkage(X_scaled, method="ward")

# Dendrogram
plt.figure(figsize=(12, 6))
dendrogram(Z, labels=df["Country"].values, leaf_rotation=90)
plt.xlabel("Country")
plt.ylabel("Distance (Ward)")
plt.title("Hierarchical Clustering Dendrogram (Ward)")
plt.tight_layout()
plt.show()
```



The K-Means solution indicated a number of 3 clusters. In order to obtain the same solution using the dendrogram, we must cut the tree just below the second to last merge, drawing a horizontal line spanning across the dendrogram at a Ward distance level (y-axis level) of approximately 11.0. This suggests the same number of 3 clusters, as a horizontal line in the described position would cross three vertical lines in the diagram. In terms of the dendrogram alone, the largest and most notable “gap” occurs in the final merge, indicating that the final two clusters remaining are very dissimilar relative to the differences that exist between previously merged groups. Thus, the hierarchical clustering suggests that there are two natural clusters, which is similar to the result received using the silhouette score. As touched on previously, this would separate countries into only two groups: overall risky and overall safe in terms of institutional and economic performance. This split is too general, ignoring the important “emerging market” group identified when there are 3 clusters.

Task #3(b) Cutting the Dendrogram: This section cuts the dendrogram such that there are 3 clusters, the resulting number from the K-Means analysis. The goal is to understand the difference

in cluster country makeup between the Ward linkage method and the previously used K-Means method by comparing the cluster assignments.

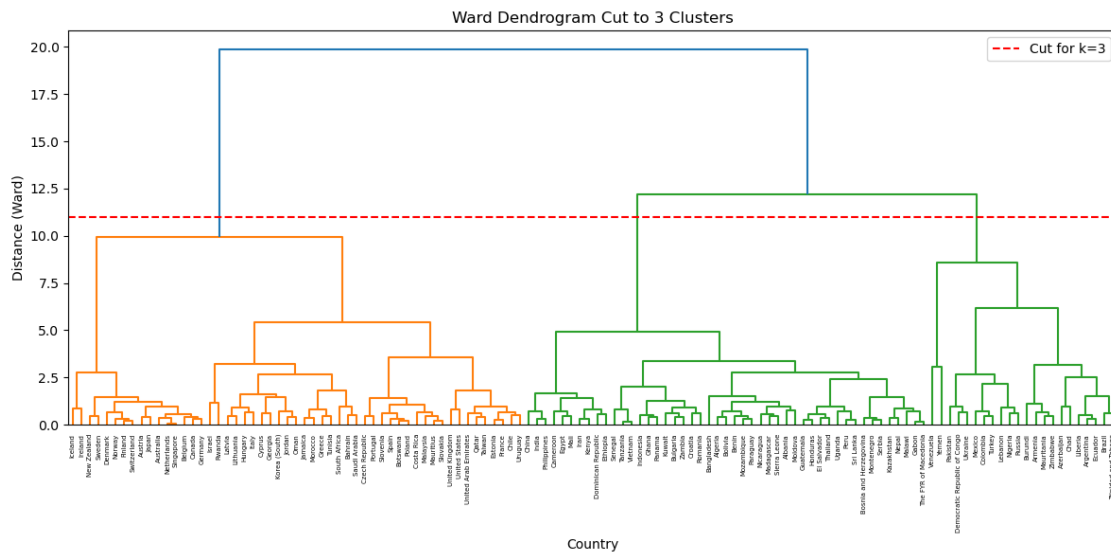
```
[3]: from scipy.cluster.hierarchy import dendrogram, fcluster
import matplotlib.pyplot as plt

# same K as your K-Means result
k = 3

# cut the hierarchy into k clusters
ward_labels = fcluster(Z, t=k, criterion='maxclust')

# draw dendrogram + cut line
cut_height = 11.0

plt.figure(figsize=(12, 6))
dendrogram(Z, labels=df["Country"].values, leaf_rotation=90)
plt.axhline(y=cut_height, color="red", linestyle="--", label=f"Cut for k={k}")
plt.title(f"Ward Dendrogram Cut to {k} Clusters")
plt.xlabel("Country")
plt.ylabel("Distance (Ward)")
plt.legend()
plt.tight_layout()
plt.show()
```



As described in section #3(b), to get a result of three clusters we must cut right below the second last merge; in this case, placing the horizontal line at a distance (ward) of 11.0 (extending along the y-axis) assigns the countries into three different clusters, which is the number suggested by the K-Means analysis.

```
[4]: from sklearn.cluster import KMeans
from scipy.optimize import linear_sum_assignment
import pandas as pd

# K-Means labels (k=3)
kmeans = KMeans(n_clusters=3, random_state=42, n_init=50)
kmeans_labels = kmeans.fit_predict(X_scaled)

# Ward labels already created from your cut at 11.0:
# ward_labels = fcluster(Z, t=11.0, criterion='distance')

# align Ward label IDs to K-Means label IDs
ward0 = ward_labels - 1
ct = pd.crosstab(kmeans_labels, ward0).reindex(index=[0,1,2], columns=[0,1,2],
    ↪fill_value=0)
r, c = linear_sum_assignment(-ct.values)
mapping = {ward_id: km_id for km_id, ward_id in zip(r, c)}
ward_aligned = pd.Series(ward0).map(mapping).to_numpy()

# count same assignments
same_count = (kmeans_labels == ward_aligned).sum()
total = len(df)

print(f"Same-cluster assignments: {same_count}/{total}")
```

Same-cluster assignments: 100/122

The K-Means analysis and the agglomerative hierarchal clustering using Ward's method are expected to give similar results because the way Ward measures distances between each pairwise object is by choosing the merge that minimizes the increase in total within-cluster variance, which is similar to the K-Means approach in the sense that K-Means minimizes the within-cluster sum of squares. However, the results will not be identical given the differences in the processes, which will be discussed in the following sections. Observing the 3 cluster assignments, we find that 100 out of the 122 total countries were assigned to the same cluster by both methods, indicating that 22 countries were placed into different clusters.

```
[5]: # countries assigned to different clusters
comparison = pd.DataFrame({
    "Country": df["Country"],
    "KMeans_Cluster": kmeans_labels,
    "Ward_Cluster": ward_aligned
})

different_countries = comparison[comparison["KMeans_Cluster"] !=
    ↪comparison["Ward_Cluster"]]
different_countries = different_countries.sort_values("Country")

print(f"Number assigned differently: {len(different_countries)}")
```

```
display(different_countries)
```

Number assigned differently: 22

	Country	KMeans_Cluster	Ward_Cluster
3	Armenia	1	2
7	Bahrain	1	0
22	Colombia	1	2
25	Cyprus	1	0
38	Georgia	1	0
41	Greece	1	0
50	Israel	1	0
51	Italy	1	0
52	Jamaica	1	0
54	Jordan	1	0
59	Latvia	1	0
61	Liberia	1	2
67	Mauritania	1	2
69	Mexico	1	2
72	Morocco	1	0
80	Oman	1	0
81	Pakistan	1	2
91	Rwanda	1	0
92	Saudi Arabia	1	0
99	South Africa	1	0
109	Tunisia	1	0
110	Turkey	1	2

The 22 countries listed above are those that are assigned to different clusters by the K-Means and Ward clusters. Using both methods is wise, as it allows these “borderline” countries to be examined. In problem #2, the three clusters were described in the following manner: Cluster 0 includes the typical “emerging markets” countries characterized by moderate insitutional risk but high GDP growth, Cluster 1 includes the “safe markets” characterized by low insitutional risk and high transparency as well as moderate GDP growth, and Cluster 2 includes the “risky markets” with systemic instability and economic contraction. It is intuitive that there are countries on the borderline of these three clusters, for example those moving from “risky market” into “emerging market” status thanks to increased indusrialization and investor confidence, but perhaps still lagging institutional systems. This could include countries such as Columbia and Turkey which display emerging-market growth potential but have persistent insitutional risk including corruption and challenges with their rule of law. Additionally, it is also intuitive that there exist countries transitioning from “emerging markets” into “safe markets”, as their large GDP growth from initial emergeance begins slowing down. This likely includes countries such as Cyprus, Latvia, and Georgia whose GDP growth has moderated to steadier rates, but whos insitutions are still not of highest quality.

Problem #3(c) Discussion: This section discusses the tradeoffs between K-Means and hierar-chical clustering to reflect on the benefits and drawbacks of both. It also touches on which might lead to the better solution in this application.

K-Means clustering requires K, the number of clusters, to be specified up front, before running the

method, whereas the hierarchical clustering instead displays a full tree structure that showcases the relationships between the observations, and allows the number of clusters to be chosen based on the natural relationships displayed. In this sense, the hierarchical method is useful when you want to explore the cluster structure, while the K-Means works well when you have a sense of the data you are working with and can reasonably use tools such as the elbow method to obtain a strong idea of K given the context. Furthermore, there exist differences in computational speed, with K-Means being the faster option as it does not require computing all pairwise distances like the hierarchical procedure. This becomes more relevant with large datasets exceeding 1000 observations, which is not extremely relevant to the country analysis completed in this lab. However, note that the K-Means method is not deterministic, and hence even with a large data set, must be run multiple times, which reduces the speed, although not significantly. Finally, the clusters are assumed to have a spherical shape in K-Means, whereas there is more flexibility with the hierarchical approach because there are different ways to calculate linkage distances, although the Ward methods results used here are most similar to K-Means. Ultimately, in this case, I agree with completing both in order to have a sanity check. There is both enough contextual information available to understand a reasonable K and a small enough dataset such that it is possible to conduct hierarchical clustering to come to two strong findings that can be compared to extract useful information on borderline cases.

0.1.1 Problem #4

This problem discusses the interpretation and investment implications of the findings from the previous sections of this lab.

Problem #4(a) Risk Ranking: This section uses the clustering analysis to create a risk ranking of the clusters from lowest risk to highest risk, justifying the ranking by referring to the cluster centers.

```
[6]: from sklearn.cluster import KMeans
import pandas as pd
# Fit the K-Means model with K=3
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
kmeans.fit(X_scaled)
# Extract the cluster centers
centroids = kmeans.cluster_centers_
# Create the table with one row per cluster and one column per feature
features = ['Corruption', 'Peace', 'Legal', 'GDP Growth']
centroids_df = pd.DataFrame(centroids, columns=features)
centroids_df.index.name = 'Cluster'
print("--- Cluster Centers (Standardized Units) ---")
display(centroids_df.round(4))
```

```
--- Cluster Centers (Standardized Units) ---
```

	Corruption	Peace	Legal	GDP Growth
Cluster				
0	-0.4796	0.2633	-0.4351	0.3759
1	1.2314	-0.9792	1.1660	0.0062
2	-1.0275	1.3522	-1.0590	-1.7201

The standardized values for corruption, peace, legal, and GDP growth factors make the cluster centers easier to interpret, since each value reports how far a cluster deviates from the overall mean (in standard deviation units), which makes clustering and comparison possible. As described in problem #2, a positive value indicates that a factor's reported level is above its mean level, while a negative value indicates that it is below its mean. A low risk center would be characterized by a low peace score, but a high corruption score, legal score, and GDP growth score. This may sound counterintuitive in terms of the peace and corruption factors, but is explained by the fact that the corruption score ranges from highly corrupt to very clean, and the peace score ranges from very peaceful to not at all peaceful. By this rule, cluster 1's center displays the closest characteristics of a perfectly low risk expectation, as its peace score is the lowest while corruption and legal are both the highest among each cluster, demonstrating high peace, transparency, and a strong legal system; but, its GDP growth is not the highest, although still above average (i.e. characteristics of a safe, stable growth market). On the other hand, cluster 2 displays the opposite characteristics of a safe market, instead suggesting its of highest risk. Not only is there low peace, high corruption, and bad legal systems, it also has the worst GDP growth, making it the most risky cluster center in all regards, that is, both institutionally and economically (i.e. risky market). Finally, cluster 0 lies in the middle of every range, except for GDP growth in which it has the highest performance (i.e. emerging market behaviour). This analysis allows us to conclude that the risk ranking of the clusters from lowest risk to highest risk is the following: Cluster 1, 0, 2.

Problem #4(b) Advising and Institutional Investor: This section finally pulls together the findings to advise an institutional investor who wants to diversify their emerging markets portfolio using the clustering results. Further, the limitations of relying on the four indicators in this analysis are discussed.

An institutional investor wishing to diversify their emerging markets portfolio can use the cluster assignments to diversify across risk profiles. An investor holding an emerging market portfolio likely has exposure similar to countries in Cluster 0, where moderate corruption, peace, and legal risk is paired with relatively high GDP growth. To diversify this exposure, they can seek exposure to countries in Clusters 1 and 2, since each cluster reflects a different risk return profile. This allows the investor to balance expected return and downside risk in a way that fits their objectives. For example, a pension fund needs both sufficient long-run returns and enough stability to support regular payouts. In that case, it may place a larger weight on the more institutionally stable markets, Cluster 1, while keeping smaller allocations to the riskier clusters for upside opportunities. The exact allocation should depend on the type of institution and their corresponding mandate, but the core idea is to use the clusters to spread exposure across different market types. At the same time, there are important limitations to relying only on these four indicators. While the clustering gives intuitive groupings, country risk depends on many additional factors that are not included here, such as currency risk, sovereign debt levels, and market liquidity and efficiency. The model gives us a classification at a high level, but it does not capture all relevant risks among countries. In addition, with only four inputs, results are very sensitive to the quality of the data; indicators like corruption and peace can contain subjectivity and may not fully capture country level nuances. Overall, the clusters are most useful as a portfolio allocation tool for improving diversification across market types, but they should not be used as a standalone decision rule, and should be combined with market knowledge and analysis before final investment decisions are made. Furthermore, there are practical constraints in gaining exposure to the different markets, including market and financial product access, but the clusters give a helpful classification for investors to refer to.